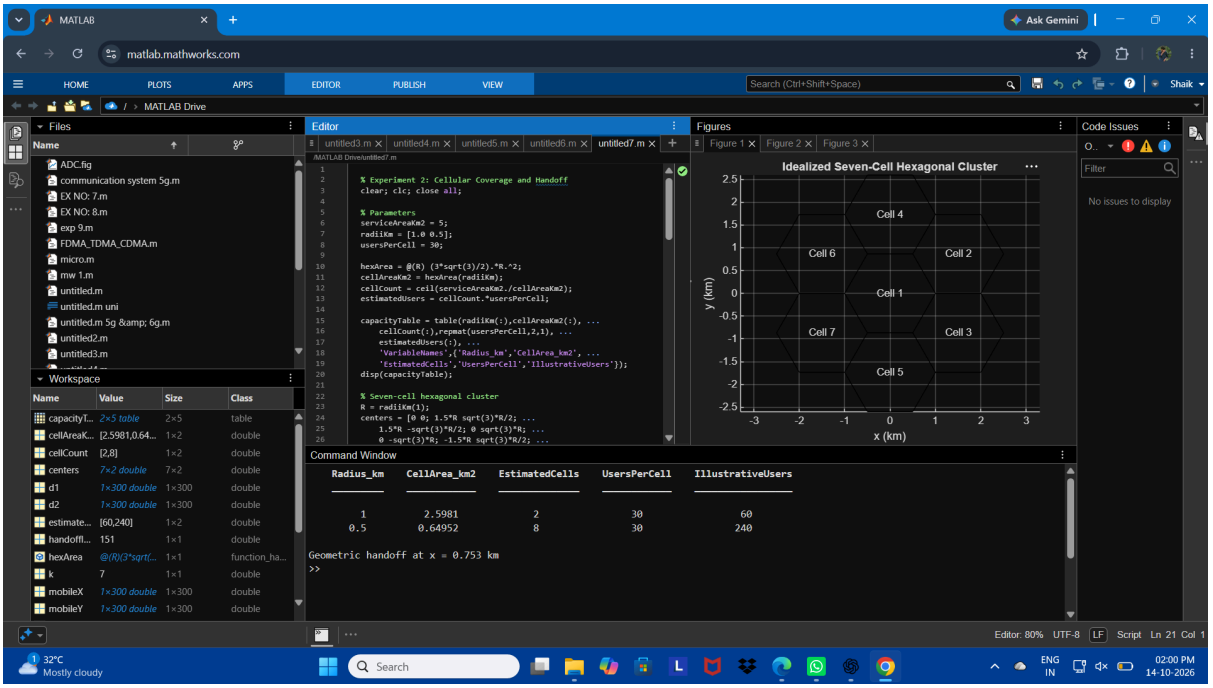
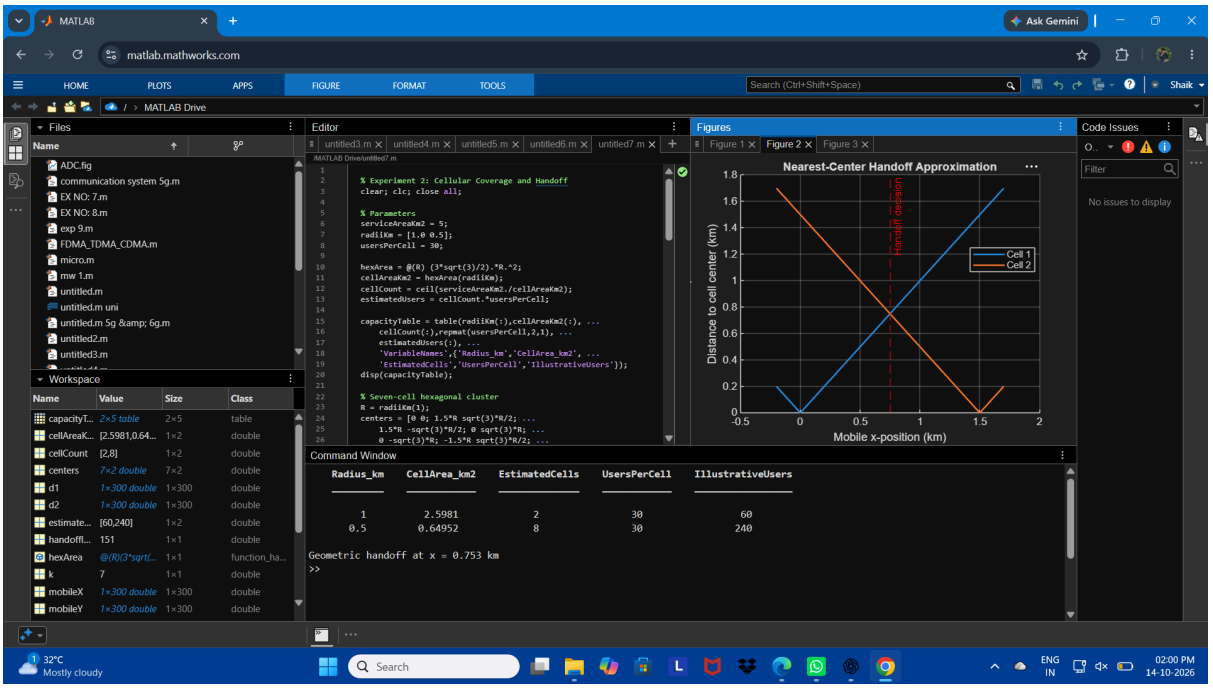
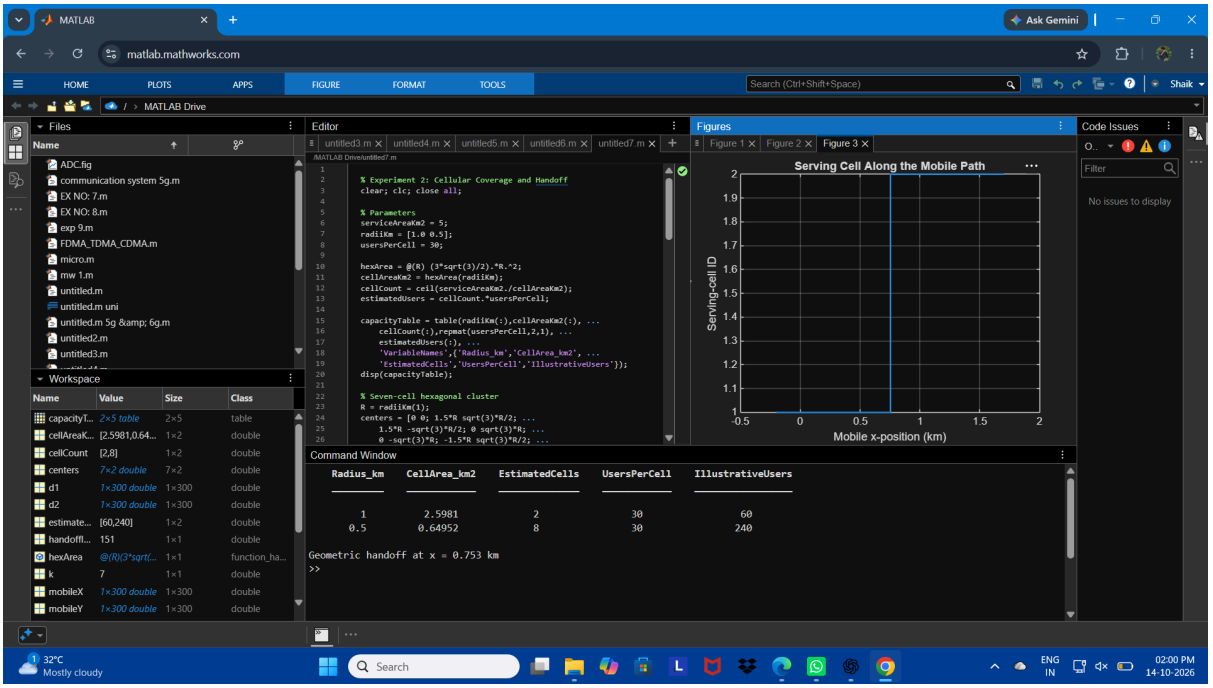
OBJECTIVE 1



OBJECTIVE 2



OBJECTIVE 3


MATLAB
matlab.mathworks.com
Ask Gemini
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Shaik
MATLAB Drive
Files
Name
ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name
Value
Size
Class
capacityT...
2×5 table
2×5
table
cellAreaK...
[2.5981,0.64...
1×2
double
cellCount
[2,8]
1×2
double
centers
7×2 double
7×2
double
d1
1×300 double
1×300
double
d2
1×300 double
1×300
double
estimate...
[60,240]
1×2
double
handoffl...
151
1×1
double
hexArea
@(R)3*sqrt(...
1×1
function_ha...
k
7
1×1
double
mobileX
1×300 double
1×300
double
mobileY
1×300 double
1×300
double
Editor
untitled3.m
untitled4.m
untitled5.m
untitled6.m
untitled7.m
% Experiment 2: Cellular Coverage and Handoff
clear; clc; close all;
% Parameters
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2).*R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2./cellAreaKm2);
estimatedUsers = cellCount.*usersPerCell;
capacityTable = table(radiiKm(:),cellAreaKm2(:), ...
cellCount(:),repmat(usersPerCell,2,1), ...
estimatedUsers(:), ...
'VariableNames',{'Radius_km','CellArea_km2', ...
'EstimatedCells','UsersPerCell','IllustrativeUsers'});
disp(capacityTable);
% Seven-cell hexagonal cluster
R = radiiKm(1);
centers = [0 0; 1.5*R sqrt(3)*R/2; ...
1.5*R -sqrt(3)*R/2; 0 sqrt(3)*R; ...
0 -sqrt(3)*R; -1.5*R sqrt(3)*R/2; ...
Figures
Figure 1
Figure 2
Figure 3
Serving Cell Along the Mobile Path
Serving-cell ID
Mobile x-position (km)
Code Issues
No issues to display
Command Window
Radius_km
CellArea_km2
EstimatedCells
UsersPerCell
IllustrativeUsers
1
2.5981
2
30
60
0.5
0.64952
8
30
240
Geometric handoff at x = 0.753 km
>>
32°C
Mostly cloudy
Search
ENG
IN
02:00 PM
14-10-2026